

# PEROTTINO Tony

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## EDUCATIONAL HISTORY

- Double Bachelor's Degree in Mathematics and Computer Science, University of Toulouse** 2023 — 2026
- Intensive academic program with an increased workload compared to standard degrees (around +30% ECTS)
  - Core Coursework: Software Engineering (Design, OOP, Architecture), CS Foundations (Algorithms, Data Structures, Networks), and Mathematics (Linear Algebra, Probability with applications in ML and AI)
- General Baccalaureate – Highest Honors (Très Bien), Lycée Marie-Louise Dissard Française** 2021 — 2023
- Specialisation in Mathematics, Physics & Chemistry. “Expert Mathematics” elective.

## PROFESSIONAL EXPERIENCE

- Software Engineering Intern, IRIT (Computer Science Research Institute of Toulouse)** 3 weeks, 2025
- Enhanced a database visualization application using Java (Spring); managed back-end tasks (database management via Hibernate, XLSX/XML/SVG processing via Apache POI) and front-end development (Thymeleaf, Bootstrap). Immersed in complex legacy code to debug, simplify, and refactor existing modules, ensuring the scalability and continuity of new features within the application's architecture.

## ACADEMIC PROJECTS

- Advanced AI & Computer Vision (Pose Estimation & Generative Models)** 2025-2026
- Conducted a comparative study of NLF and MMPose architectures to generate a Digital Twin of human hand motion
  - Computer Vision Workflows: Architected an automated pipeline using Segment Anything (SAM) and OpenCV; specialized in the conversion of raw segmentation data into isolated RGBA assets for professional-grade composition.
  - Subject-Consistent Synthesis: Implemented generative inpainting solutions; engineered specific prompt-masking strategies to preserve edge fidelity and subject integrity during large-scale background replacement.
- Stochastic Optimization Research (Hastings-Metropolis Algorithm) [\[link\]](#)** 2025
- Authored a 40-page research report focused on the mathematical theory of stochastic processes; applied the Hastings-Metropolis algorithm to optimize solutions for the Traveling Salesman Problem (TSP).
- Piano Analytics (MMPose & MIDI) [\[link\]](#)** 2025
- Developed a spatial mapping algorithm to synchronize MMPose hand-tracking with MIDI data, enabling visual-only note estimation and pressed-key analysis while evaluating model robustness against motion blur and finger occlusions.
- Energy Consumption Forecasting (Enedis Hackathon) [\[link\]](#)** 2025
- Participated in a competitive Hackathon to predict power grid consumption using LSTM Neural Networks, Ridge regression, and Gradient Boosting. Achieved the top performance in the class and ranked 12th out of 118 on the global leaderboard.
- Systems & Network Engineering (C & TCP/UDP Implementation)** 2024
- Implemented TCP/UDP network layers and developed complex data structures including Red-Black Trees in C with Graphviz visualization.
- In relation with Software Engineering (Java & Python AI)** 2024
- Designed a restaurant reservation interface in Java ; engineered a board game engine featuring a Minimax-based AI algorithm in Python.

## PERSONAL PROJECTS

- Physics Simulation & Content Automation (Godot/GDScript)** 2025
- Engineered a 2D physics simulation engine using Godot (GDScript); developed an automated pipeline to generate and render social media video ([TikTok](#)) content based on simulation data.
- Game Development & Algorithm Design** 2019 – 2024
- Implemented the [Ultimate Tic Tac Toe](#) variant using OOP principles in Godot; developed custom AI opponents based for instance on decision-tree logic and heatmaps.
  - Designed a logic-based mobile puzzle game prototype in Unity (C#), handling game design, scripting, and music composition.
- Challenges about computer science & maths**
- Computer Science: Participant in [Advent of Code 2024](#) or [LeetCode](#) (~100 problems) ; Mathematics: participation in [competitions](#) and Olympiads

## SKILLS

**Technical:** Java (Spring, Hibernate), Python, C, C#, JavaScript, OCaml, SQL, GDScript, Git, LaTeX, Apache POI, Bootstrap, HTML  
**Languages:** French (Native), English (B2), German (A2/B1)